

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Power Electronics	Course Code:	EE3032
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	20-February 2025	Total Marks:	30
	Section:	BEE-6A	Weight:	1
	Exam Type:	Assignment- 2	Page(s):	1
			CLO #	2

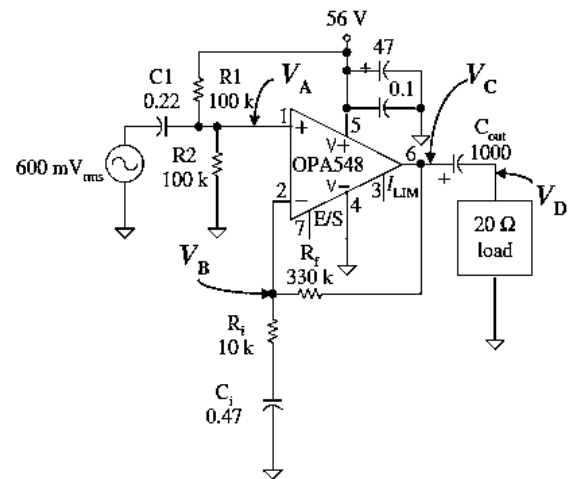
Student Name: _____ **Roll No.** _____ **Section:** _____

- Instructions:**
1. Use A4-sized sheets for this assignment.
 2. Avoid cutting and overwriting.
 3. Secure the pages with staple, clip, or binding. Do not submit loose pages.
 4. Assignment must be submitted on or before deadline during office hours only.
 5. Late submission will incur a penalty of 20% per day. After 2 days, assignment will no longer be accepted.

Question No. 1 (CLO No. 2)

Marks: 10

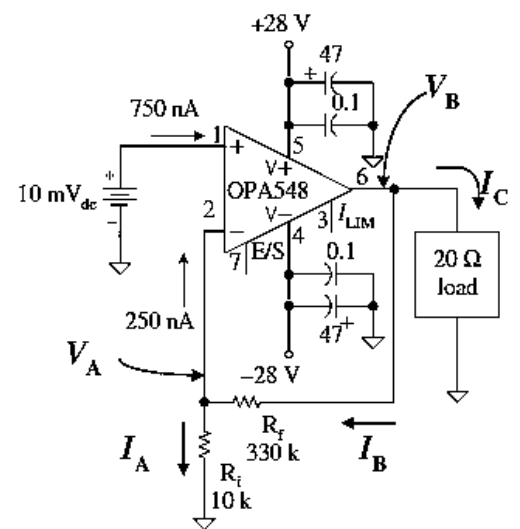
Analyze the audio amplifier shown below and calculate the dc and rms values of each of the four indicated voltages.



Question No. 2 (CLO No. 2)

Marks: 10

Considering the effect of non-ideal input offset voltage, bias currents, and offset currents; analyze the following audio amplifier and calculate all the indicated voltages and currents. (Hint: Start calculations in the mentioned order)



Question No. 3 (CLO No. 2)

Marks: 10

Analyze the power dissipation of an LM386 audio amplifier operating with a 12 V power supply while delivering 0.5 W into an 8 Ω load. Determine the power supplied by the source and the power dissipated by the device.

Q. No. 1DC Values

$$V_B = \frac{10k}{10k + 330k} (56)$$

$$= 1.6470 \text{ V}$$

$$A_B \quad V_A = V_B = 1.6470 \text{ V} \quad (\text{virtual short circuit})$$

$$\Rightarrow V_O = 0 \quad (\text{capacitor blocks DC voltage})$$

$\Rightarrow$ Non-inverting configuration

$$A \quad G = 1 + \frac{R_F}{R_i} = 1 + \frac{330k}{10k}$$

$$\Rightarrow G = \frac{V_O}{V_i} = 34$$

$$\Rightarrow V_O = V_C = (34) V_i$$

$$= (34)(600 \text{ mV})$$

$$V_C = 20.4 \text{ V}$$

RMS Value

$$A \quad G = 1 + \frac{R_F}{R_i} = 1 + \frac{330k}{10k} = 34$$

$$\Rightarrow G = \frac{V_O}{V_i} = 34$$

$$V_O = (34)(600 \text{ mV})$$

$$\Rightarrow V_C = 20.4 \text{ V}$$

$$\Rightarrow V_A = 600 \text{ mV} \quad (\because \text{from supply})$$

$$\Rightarrow V_B = V_A = 600 \text{ mV} \quad (\text{virtual short circuit})$$

Bee

$\Rightarrow$ Capacitor blocks DC voltage

$$\Rightarrow V_O = V_C = 20.4 \text{ V}$$

Q.1 No.2

$$\Rightarrow V_+ = 10\text{mV}$$

$$\Rightarrow V_A = V_+ = 10\text{mV}$$

$$\Rightarrow I_A = \frac{V_A}{R_i} = \frac{10\text{m}}{10\text{k}} = 1\mu\text{A}$$

$$\Rightarrow G = 1 + \frac{R_f}{R_i} = 1 + \frac{330\text{k}}{10\text{k}} = 34$$

$$\Rightarrow G = \frac{V_o}{V_+} = 34$$

$$V_o = (34)(10\text{m})$$

$$\Rightarrow V_B = 0.34\text{V}$$

$$\Rightarrow I_B = I_A + 250\text{n}$$
$$= 1\mu + 250\text{n}$$

$$I_B = 1.25\mu\text{A}$$

$$\Rightarrow \boxed{V_B - 0 = I_C R_L}$$

$$I_C = \frac{0.34}{20}$$

$$\Rightarrow I_C = 17\text{mA}$$

Q.1 No.3

$$\Rightarrow P_L = \frac{V_o^2}{R_L}$$

$$\begin{aligned}\Rightarrow V_o &= \sqrt{P_L R_L} \\ (\text{rms}) &= \sqrt{(0.5)(8)} \\ &= 2 \text{ V}\end{aligned}$$

$$\Rightarrow V_{o(\text{p})} = (2)(\sqrt{2}) = 2.83 \text{ V}$$

$$\Rightarrow P_L = I_L^2 R_L$$

$$\begin{aligned}\Rightarrow I_L &= \sqrt{\frac{P_L}{R_L}} \\ &= \sqrt{\frac{0.5}{8}} \\ I_L &= 0.25 \text{ A}\end{aligned}$$

$$\begin{aligned}\Rightarrow P_{\text{supply}} &= V_s I_L \\ &= (12)(0.25) \\ &= 3\end{aligned}$$

$$\begin{aligned}\Rightarrow P_{\text{IC}} &= P_{\text{supply}} - P_L \\ &= 3 - 0.5 \\ &= 2.5 \text{ W}\end{aligned}$$